

数值算法 Homework 09

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Problem 1

Show that the n -point Gauss-Chebyshev quadrature rule reads:

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx \approx \frac{\pi}{n} f\left(\cos\left(\frac{(2k-1)\pi}{2n}\right)\right)$$

Solution:

当权函数 $\rho(x) = \frac{1}{\sqrt{1-x^2}}$ 且积分区间为 $[-1, 1]$ 时, 相应的正交多项式为 Chebyshev 多项式:

$$T_n(x) := \cos[n \arccos(x)] = \frac{(x + \sqrt{x^2 - 1})^n + (x - \sqrt{x^2 - 1})^n}{2}$$
$$\int_{-1}^1 \frac{1}{\sqrt{1-x^2}} T_m(x) T_n(x) dx = \begin{cases} 0, & m \neq n \\ \frac{\pi}{2}, & m = n \neq 0 \\ \pi, & m = n = 0 \end{cases}$$

记 n 次 Chebyshev 多项式的 n 个根为 $x_1, x_2, \dots, x_n$ (其中 $x_k = \cos(\frac{2k-1}{2n}\pi)$)

则 $T_n(x) = 2^{n-1} \prod_{k=1}^n (x - x_k) = 2^{n-1} \omega_n(x)$ (因为其首项系数为 2^{n-1})

因此其求积系数为:

$$\begin{aligned} A_k &= \int_{-1}^1 \rho(x) l_k(x) dx \\ &= \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} \prod_{\substack{i=1 \\ i \neq k}}^n \frac{x - x_i}{x_k - x_i} dx \\ &= \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} \cdot \frac{\prod_{i=1}^n (x - x_i)}{(x - x_k) \prod_{i=1, i \neq k}^n (x_k - x_i)} dx \quad (\text{note that } T'_n(x_k) = 2^{n-1} \prod_{\substack{i=1 \\ i \neq k}}^n (x_k - x_i)) \\ &= \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} \cdot \frac{T_n(x)}{(x - x_k) T'_n(x_k)} dx \end{aligned}$$

根据 Chebyshev 多项式的递推关系 $T_n(x) = 2xT_{n-1}(x) - T_{n-2}(x)$ 易知:

$$\begin{aligned} &T_n(x)T_{n-1}(y) - T_n(y)T_{n-1}(x) \\ &= [2xT_{n-1}(x) - T_{n-2}(x)]T_{n-1}(y) - [2yT_{n-1}(y) - T_{n-2}(y)]T_{n-1}(x) \\ &= 2(x-y)T_{n-1}(x)T_{n-1}(y) + [T_{n-1}(x)T_{n-2}(y) - T_{n-1}(y)T_{n-2}(x)] \\ &= \dots \\ &= 2(x-y) \sum_{i=1}^{n-1} T_i(x)T_i(y) + [T_1(x)T_0(y) - T_1(y)T_0(x)] \\ &= 2(x-y) \sum_{i=1}^{n-1} T_i(x)T_i(y) + [x \cdot 1 - y \cdot 1] \\ &= 2(x-y) \left(\frac{1}{2} + \sum_{i=1}^{n-1} T_i(x)T_i(y) \right) \end{aligned}$$

令 $y = x_k$, 由 $T_n(x_k) = 0$ 可得:

$$\begin{aligned}
T_n(x)T_{n-1}(x_k) - T_n(x_k)T_{n-1}(x) &= 2(x - x_k) \left(\frac{1}{2} + \sum_{i=1}^{n-1} T_i(x)T_i(x_k) \right) \\
&\Updownarrow \\
T_n(x)T_{n-1}(x_k) &= 2(x - x_k) \left(\frac{1}{2} + \sum_{i=1}^{n-1} T_i(x)T_i(x_k) \right) \\
&\Updownarrow \\
\frac{T_n(x)}{(x - x_k)T'_n(x_k)} &= \frac{2}{T_{n-1}(x_k)T'_n(x_k)} \left(\frac{1}{2} + \sum_{i=1}^{n-1} T_i(x)T_i(x_k) \right) \\
&= \frac{1}{T_{n-1}(x_k)T'_n(x_k)} + \frac{2}{T_{n-1}(x_k)T'_n(x_k)} \sum_{i=1}^{n-1} T_i(x)T_i(x_k)
\end{aligned}$$

代入 A_k 的表达式即有:

$$\begin{aligned}
A_k &= \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} \cdot \frac{T_n(x)}{(x - x_k)T'_n(x_k)} dx \\
&= \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} \cdot \left\{ \frac{1}{T_{n-1}(x_k)T'_n(x_k)} + \frac{2}{T_{n-1}(x_k)T'_n(x_k)} \sum_{i=1}^{n-1} T_i(x)T_i(x_k) \right\} dx \\
&= \frac{1}{T_{n-1}(x_k)T'_n(x_k)} \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx + 2 \sum_{i=1}^{n-1} \frac{T_i(x_k)}{T_{n-1}(x_k)T'_n(x_k)} \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} T_i(x) dx \\
&\quad \left(\text{note that } T_i(x) \ (n \geq 1) \text{ is orthogonal to } T_0(x) \equiv 1, \text{ i.e. } \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} T_i(x) dx = 0 \right) \\
&= \frac{1}{T_{n-1}(x_k)T'_n(x_k)} \arcsin(x) \Big|_{-1}^1 + 2 \sum_{i=1}^{n-1} \frac{T_i(x_k)}{T_{n-1}(x_k)T'_n(x_k)} \cdot 0 \\
&= \frac{\pi}{T_{n-1}(x_k)T'_n(x_k)}
\end{aligned}$$

根据定义 $T_n(x) = \cos[n \arccos(x)] = \cos(n\theta)$ (其中 $x = \cos(\theta)$)

因此我们有:

$$\begin{aligned}
T'_n(x) &= \frac{d}{d\theta} \cos(n\theta) \cdot \frac{d\theta}{dx} \\
&= -n \sin(n\theta) \cdot \frac{1}{-\sin(\theta)} \\
&= \frac{n \sin(n\theta)}{\sin(\theta)}
\end{aligned}$$

注意到 $x_k = \cos(\frac{2k-1}{2n}\pi)$ 对应 $\theta_k = \frac{2k-1}{2n}\pi$, 于是我们有:

$$\begin{aligned}
A_k &= \frac{\pi}{T_{n-1}(x_k)T'_n(x_k)} \\
&= \frac{\pi \cdot \sin(\theta_k)}{\cos((n-1)\theta_k) \cdot [n \sin(n\theta_k)]} \\
&= \frac{\pi \sin(\theta_k)}{[\cos(n\theta_k) \cos(\theta_k) + \sin(n\theta_k) \sin(\theta_k)] \cdot [n \sin(n\theta_k)]} \\
&\quad \left(\text{note that } \begin{cases} \sin(n\theta_k) = \sin(n \cdot \frac{2k-1}{2n} \pi) = \sin(\frac{2k-1}{2} \pi) = (-1)^{k+1} \\ \cos(n\theta_k) = \cos(n \cdot \frac{2k-1}{2n} \pi) = \cos(\frac{2k-1}{2} \pi) = 0 \end{cases} \right) \\
&= \frac{\pi \sin(\theta_k)}{[0 \cdot \cos(\theta_k) + (-1)^{k+1} \cdot \sin(\theta_k)] \cdot [n \cdot (-1)^{k+1}]} \\
&= \frac{\pi}{n}
\end{aligned}$$

因此 Gauss-Chebyshev 求积公式为:

$$\int_{-1}^1 \frac{1}{\sqrt{1-x^2}} f(x) dx \approx \sum_{k=1}^n A_k f(x_k) = \frac{\pi}{n} \sum_{k=1}^n f\left(\cos\left(\frac{2k-1}{2n}\pi\right)\right)$$

其截断误差为:

$$\begin{aligned}
E_n(f) &= \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} f(x) dx - \frac{\pi}{n} \sum_{k=1}^n f\left(\cos\left(\frac{2k-1}{2n}\pi\right)\right) \\
&= \frac{\pi}{2^{2n-1}(2n-1)!} f^{(2n)}(\eta) \text{ for some } \eta \in (-1, 1)
\end{aligned}$$

Problem 2

(Filon Quadrature)

Develop a quadrature rule for the integral $\int_a^b \cos(mx) f(x) dx$

such that it provides exact results for polynomials of degree up to three.

- **(数值逼近, 5.3.3 节, 例 3)** 待定系数法确定 Gauss 型求积公式的节点和系数
- **待纠正:** $\cos(mx)$ 不一定恒正!

Solution:

我们记:

$$\begin{aligned}
I_0 &= \int_a^b \cos(mx) dx \\
&= \frac{\sin(mb) - \sin(ma)}{m}
\end{aligned}$$

$$\begin{aligned}
I_1 &= \int_a^b x \cos(mx) dx \\
&= \frac{b \sin(mb) - a \sin(ma)}{m} + \frac{\cos(mb) - \cos(ma)}{m^2}
\end{aligned}$$

$$\begin{aligned}
I_2 &= \int_a^b x^2 \cos(mx) dx \\
&= \frac{b^2 \sin(mb) - a^2 \sin(ma)}{m} + \frac{2(b \cos(mb) - a \cos(ma))}{m^2} - \frac{2(\sin(mb) - \sin(ma))}{m^3}
\end{aligned}$$

$$\begin{aligned}
I_3 &= \int_a^b x^3 \cos(mx) dx \\
&= \frac{b^3 \sin(mb) - a^3 \sin(ma)}{m} + \frac{3(b^2 \cos(mb) - a^2 \cos(ma))}{m^2} - \frac{6(b \sin(mb) - a \sin(ma))}{m^3} - \frac{6(\cos(mb) - \cos(ma))}{m^4}
\end{aligned}$$

要使形如 $I_1(f) := A_0x_0 + A_1x_1$ 的求积公式要具有 3 次代数精度,
其求积系数和求积节点必须满足以下方程组:

$$\begin{aligned} A_0 + A_1 &= I_0 \\ A_0x_0 + A_1x_1 &= I_1 \\ A_0x_0^2 + A_1x_1^2 &= I_2 \\ A_0x_0^3 + A_1x_1^3 &= I_3 \end{aligned}$$

我们有:

$$\begin{aligned} \begin{bmatrix} A_0 \\ A_1 \end{bmatrix} &= \begin{bmatrix} 1 & 1 \\ x_0 & x_1 \end{bmatrix}^{-1} \begin{bmatrix} I_0 \\ I_1 \end{bmatrix} = \begin{bmatrix} x_0^2 & x_1^2 \\ x_0^3 & x_1^3 \end{bmatrix}^{-1} \begin{bmatrix} I_2 \\ I_3 \end{bmatrix} \\ &\Downarrow \\ \begin{cases} I_1x_0x_1 = I_2(x_0 + x_1) - I_3 \\ I_0(x_0x_1)^2 = I_2(x_0^2 + x_0x_1 + x_1^2) - I_3(x_0 + x_1) \end{cases} \end{aligned}$$

作变量代换 $u = x_0x_1$ 和 $v = x_0 + x_1$ 可得:

$$\begin{cases} I_2v - I_3 - I_1u = 0 \\ I_2v^2 - I_3v - I_2u - I_0u^2 = 0 \end{cases}$$

解得 u, v 后进而可得 x_0, x_1, A_0, A_1 , 最后得到求积公式 $I_1(f) = A_0x_0 + A_1x_1$
这里就不继续算了.

Problem 3

Write a program to compute $\int_0^1 e^x dx$ and $\int_0^1 x^{3/2} dx$ using the Gauss-Legendre quadrature rule.
Plot the quadrature error with respect to the number of quadrature nodes.

Solution:

定义自伴线性算子 $F[p](x) := x \cdot p(x) \ (\forall x \in \mathbb{R})$

生成正交多项式基的过程是一个 Lanczos 过程:

```

Given polynomial function  $p_0(x)$  of degree zero such that  $\|p_0\| = 1$ 
 $\overline{k = -1; \beta_{-1} = 1; p_{-1} \equiv 0; r_{-1} = p_0}$ 
while  $\beta_k \neq 0$ 
     $p_{k+1} = \frac{r_k}{\beta_k}$ 
     $k = k + 1$ 
     $\alpha_k = \langle F[p_k], p_k \rangle$ 
     $r_k = F[p_k] - \alpha_k p_k - \beta_{k-1} p_{k-1}$ 
     $\beta_k = \|r_k\|$ 
end
    
```

其中 $\{p_k\}$ 和 $\{r_k\}$ 均是多项式序列, $\langle \cdot, \cdot \rangle$ 和 $\|\cdot\|$ 是函数空间上的内积和范数.
于是对于任意 $k \in \mathbb{Z}_+$ 我们都有:

$$\begin{aligned}
F[P_k] &= F[p_0, p_1, \dots, p_k] \\
&= [p_0, p_1, \dots, p_k, p_{k+1}] \begin{bmatrix} \alpha_0 & \beta_0 & & & \\ \beta_0 & \alpha_1 & & & \\ & & \ddots & & \\ & & & \ddots & \beta_{k-1} \\ & & & \beta_{k-1} & \alpha_k \\ & & & & \beta_k \end{bmatrix}_{(k+2) \times (k+1)} \\
&= P_{k+1} \tilde{T}_{k+1} \\
&= [P_k, p_{k+1}] \begin{bmatrix} T_{k+1} \\ \beta_k e_{k+1}^T \end{bmatrix} \\
&= P_k T_{k+1} + \beta_k p_{k+1} \\
&= P_k T_{k+1} + r_k
\end{aligned}$$

其中 e_{k+1} 是 $\mathbb{R}^{k+1}$ 空间的第 $k+1$ 个标准正交基向量.

这样我们就得到了三项递推关系:

$$\begin{aligned}
F[p_k] &= \beta_{k-1} p_{k-1} + \alpha_k p_k + \beta_k p_{k+1} \\
\text{where } F[p_k](x) &= x \cdot p_k(x)
\end{aligned}$$

可以证明 $p_n(x)$ 有 n 个不同的根, 且它们是不可约对称三对角阵 T_{n-1} 的特征值.

Legendre 多项式是一类在区间 $[-1, 1]$ 上关于权重函数 1 正交的多项式序列:

(注意这里的 P_n 记号和上面 Lanczos 过程中的 P_n 记号重复了, 它们指代的不是同一种东西)

$$\begin{aligned}
P_n(x) &:= \frac{((x^2 - 1)^n)^{(n)}}{2^n n!} \\
\int_{-1}^1 P_m(x) P_n(x) dx &= \begin{cases} 0, & \text{if } m \neq n \\ \frac{2}{2n+1}, & \text{if } m = n \end{cases}
\end{aligned}$$

三项递推公式:

$$\begin{aligned}
P_0(x) &\equiv 1 \\
P_1(x) &= x \\
P_2(x) &= \frac{3x^2 - 1}{2} \\
P_3(x) &= \frac{5x^3 - 3x}{2} \\
\hline
(n+1)P_{n+1}(x) &= (2n+1)xP_n(x) - nP_{n-1}(x)
\end{aligned}$$

我们定义单位化的 Legendre 多项式为:

$$p_n(x) := \sqrt{\frac{2n+1}{2}} P_n(x)$$

对应的三项递推公式为:

$$\begin{aligned}
(n+1)P_{n+1}(x) &= (2n+1)xP_n(x) - nP_{n-1}(x) \\
&\Updownarrow \\
(n+1)\sqrt{\frac{2n+3}{2}}p_{n+1}(x) &= (2n+1)\sqrt{\frac{2n+1}{2}}xp_n(x) - n\sqrt{\frac{2n-1}{2}}p_{n-1}(x) \\
&\Updownarrow \\
xp_n(x) &= \frac{n+1}{\sqrt{(2n+1)(2n+3)}}p_{n+1}(x) + \frac{n}{\sqrt{(2n-1)(2n+1)}}p_{n-1}(x)
\end{aligned}$$

下面推导 α_k 和 β_k 的通项公式: (其中 $k = 0, 1, \dots$)

- ① 注意到单位化的 Legendre 多项式满足奇偶性: $p_k(-x) = (-1)^k p_k(x)$
考虑 α_k 的计算:

$$\begin{aligned}
\alpha_k &:= \langle F[p_k], p_k \rangle \\
&= \int_{-1}^1 xp_k(x)p_k(x)dx \\
&= \int_{-1}^1 x[p_k(x)]^2 dx \quad (\text{note that } x[p_k(x)]^2 \text{ is an odd function for any } k) \\
&= 0
\end{aligned}$$

- ② 单位化的 Legendre 多项式的递推关系是:

$$xp_k(x) = \frac{k+1}{\sqrt{(2k+1)(2k+3)}}p_{k+1}(x) + \frac{k}{\sqrt{(2k-1)(2k+1)}}p_{k-1}(x)$$

与 Lanczos 过程的两项递推公式 $xp_k(x) = \beta_{k-1}p_{k-1}(x) + \beta_k p_{k+1}(x)$ 比较,
(注意根据 ① 我们知道 $\alpha_k = 0$) 可得:

$$\beta_k = \frac{k+1}{\sqrt{(2k+1)(2k+3)}}$$

这样我们就可以构造不可约对称三对角阵 $T_{k-1} \in \mathbb{R}^{k \times k}$ 了,
其特征根即为单位化 Legendre 多项式的根, 也即 Legendre 多项式的根.

(Gauss-Legendre 求积公式的 Lanczos 构造法)

计算 Lanczos 系数矩阵的谱分解:

$$\begin{aligned}
\begin{bmatrix} \alpha_0 & \beta_0 & & \\ \beta_0 & \alpha_1 & \ddots & \\ & \ddots & \ddots & \beta_{n-2} \\ & & \beta_{n-2} & \alpha_{n-1} \end{bmatrix} &= \begin{bmatrix} 0 & \frac{1}{\sqrt{3}} & & \\ \frac{1}{\sqrt{3}} & 0 & \ddots & \\ & \ddots & \ddots & \frac{n-1}{\sqrt{(2n-3)(2n-1)}} \\ & & \frac{n-1}{\sqrt{(2n-3)(2n-1)}} & 0 \end{bmatrix} \\
&= \begin{bmatrix} q_{1,1} & q_{1,2} & \cdots & q_{1,n} \\ q_{2,1} & q_{2,2} & \cdots & q_{2,n} \\ \vdots & \vdots & & \vdots \\ q_{n,1} & q_{n,2} & \cdots & q_{n,n} \end{bmatrix} \begin{bmatrix} x_1 & & & \\ & x_2 & & \\ & & \ddots & \\ & & & x_n \end{bmatrix} \begin{bmatrix} q_{1,1} & q_{1,2} & \cdots & q_{1,n} \\ q_{2,1} & q_{2,2} & \cdots & q_{2,n} \\ \vdots & \vdots & & \vdots \\ q_{n,1} & q_{n,2} & \cdots & q_{n,n} \end{bmatrix}^T
\end{aligned}$$

则特征值 $x_1, x_2, \dots, x_n$ 即为 Gauss-Legendre 节点,
而 Gauss-Legendre 求积系数为:

$$\begin{aligned}
 A_k &= \frac{|q_{1,k}|^2}{p_0^2} \\
 &= |q_{1,k}|^2 \int_{-1}^1 1 dx \quad (k = 1, 2, \dots, n) \\
 &= 2|q_{1,k}|^2
 \end{aligned}$$

即可得到 n 节点的 Gauss-Legendre 求积公式:

$$\int_{-1}^1 f(x) dx \approx \sum_{k=1}^n A_k f(x_k)$$

(Gauss-Legendre 求积公式的推广)

由于权函数恒为 1, 故 Gauss-Legendre 求积公式可以通过变量代换推广到区间 $[a, b]$ 上:

$$t := \frac{b-a}{2}x + \frac{a+b}{2} \quad (x \in [-1, 1])$$

于是我们有:

$$\begin{aligned}
 \int_a^b f(t) dt &= \frac{b-a}{2} \int_{-1}^1 f\left(\frac{b-a}{2}x + \frac{a+b}{2}\right) dx \\
 &= \frac{b-a}{2} \sum_{k=1}^n A_k f\left(\frac{b-a}{2}x_k + \frac{a+b}{2}\right)
 \end{aligned}$$

因此 Gauss-Legendre 求积公式是最常用的 Gauss 型求积公式.

Matlab 实现:

```

function [nodes, weights] = gauss_legendre(n)
% 计算n点Gauss-Legendre求积的节点和权重 (区间[0,1])
if n == 0
    nodes = [];
    weights = [];
    return;
elseif n == 1
    nodes = 0.5;
    weights = 1;
    return;
end

% 构造次对角线元素 beta_k
beta = zeros(n-1, 1);
for k = 0:n-2
    beta(k+1) = (k + 1) / sqrt((2*k + 1) * (2*k + 3));
end

% 构造对称三对角矩阵 T
T = diag(beta, 1) + diag(beta, -1);

% 计算特征值和特征向量
[V, D] = eig(T);
eigenvalues = diag(D);
[eigenvalues, idx] = sort(eigenvalues);
v = V(:, idx);

```

```

% 计算权重
A_k = 2 * (V(1, :).^2)';

% 转换到区间 [0, 1]
nodes = (eigenvalues + 1) / 2;
weights = A_k * 0.5;
end

```

函数调用:

```

% 定义被积函数和精确解
f1 = @(x) exp(x);
exact1 = exp(1) - 1;

f2 = @(x) x.^(3/2);
exact2 = 2/5;

% 最大节点数
max_n = 20;

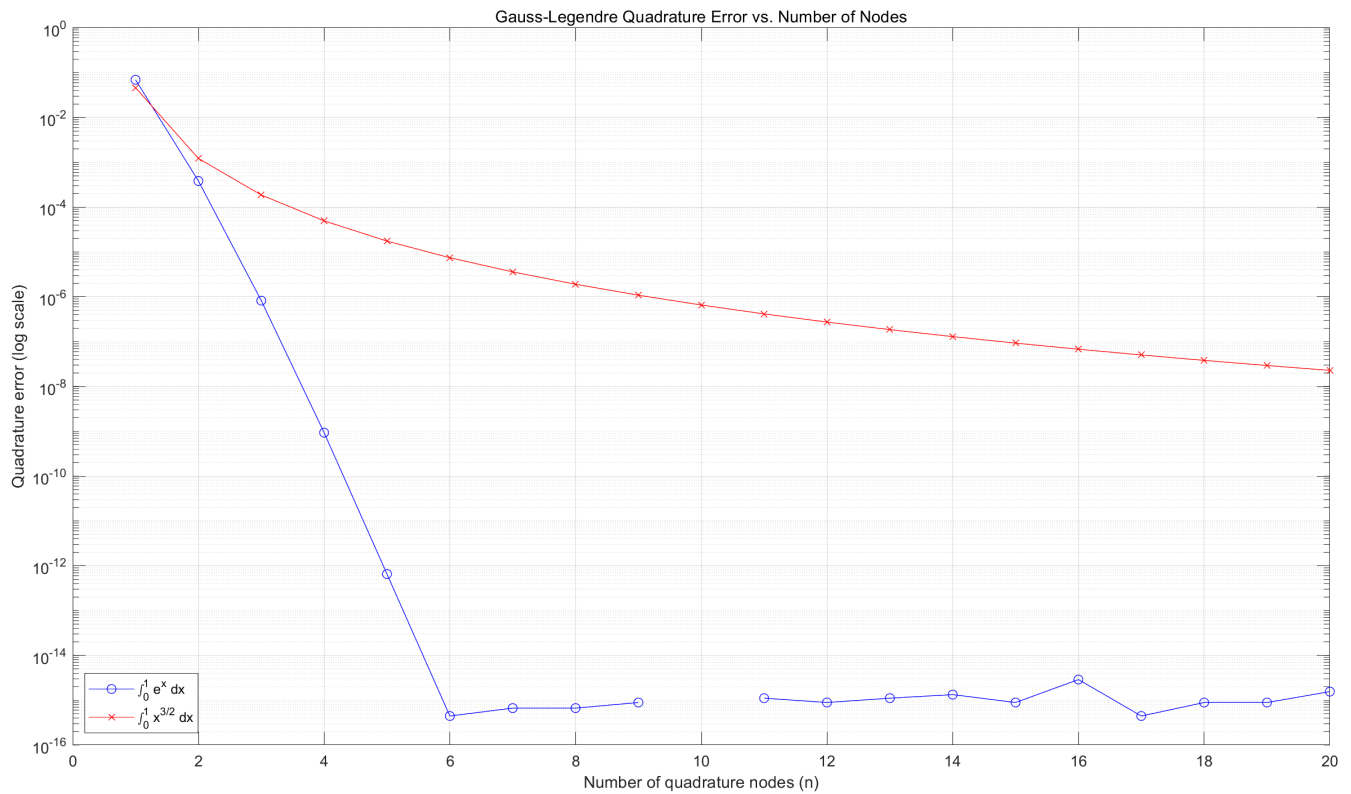
% 初始化误差数组
err1 = zeros(max_n, 1);
err2 = zeros(max_n, 1);

% 对每个n计算积分和误差
for n = 1:max_n
    [nodes, weights] = gauss_legendre(n);
    approx1 = sum(weights .* f1(nodes));
    approx2 = sum(weights .* f2(nodes));
    err1(n) = abs(approx1 - exact1);
    err2(n) = abs(approx2 - exact2);
end

% 绘制误差图
figure;
semilogy(1:max_n, err1, 'b-o', 1:max_n, err2, 'r-x');
xlabel('Number of quadrature nodes (n)');
ylabel('Quadrature error (log scale)');
legend('\int_0^1 e^x dx', '\int_0^1 x^{3/2} dx', 'Location', 'southwest');
grid on;
title('Gauss-Legendre Quadrature Error vs. Number of Nodes');

```

运行结果:



Problem 4

Determine the degree of exactness of the following 2D quadrature rule:

$$\int_0^1 \int_0^{1-y} f(x, y) dx dy \approx \frac{1}{6} \left[f\left(\frac{2}{3}, \frac{1}{6}\right) + f\left(\frac{1}{6}, \frac{2}{3}\right) + f\left(\frac{1}{6}, \frac{1}{6}\right) \right]$$

Hint: Check whether the quadrature rules provide exact results for bivariate polynomials $1, x, y, x^2, xy, y^2, x^3, x^2y, xy^2, y^3, \dots$

Solution:

$$\begin{aligned} \int_0^1 \int_0^{1-y} x^i y^j dx dy &= \int_0^1 y^j \int_0^{1-y} x^i dx dy \\ &= \int_0^1 y^j \frac{(1-y)^{i+1}}{i+1} dy \\ &= \frac{i!j!}{(i+j+2)!} \end{aligned}$$

Monomial	Exact Integral $\frac{i!j!}{(i+j+1)!}$	Quadrature Rule
1	$\frac{0!0!}{2!} = \frac{1}{2}$	$\frac{1}{6}(1+1+1) = \frac{1}{2}$
x	$\frac{1!0!}{3!} = \frac{1}{6}$	$\frac{1}{6}\left(\frac{2}{3} + \frac{1}{6} + \frac{1}{6}\right) = \frac{1}{6}$
y	$\frac{0!1!}{3!} = \frac{1}{6}$	$\frac{1}{6}\left(\frac{1}{6} + \frac{2}{3} + \frac{1}{6}\right) = \frac{1}{6}$
x^2	$\frac{2!0!}{4!} = \frac{1}{12}$	$\frac{1}{6}\left(\left(\frac{2}{3}\right)^2 + \left(\frac{1}{6}\right)^2 + \left(\frac{1}{6}\right)^2\right) = \frac{1}{12}$
xy	$\frac{1!1!}{4!} = \frac{1}{24}$	$\frac{1}{6}\left(\left(\frac{2}{3} \cdot \frac{1}{6}\right) + \left(\frac{1}{6} \cdot \frac{2}{3}\right) + \left(\frac{1}{6} \cdot \frac{1}{6}\right)\right) = \frac{1}{24}$
y^2	$\frac{0!2!}{4!} = \frac{1}{12}$	$\frac{1}{6}\left(\left(\frac{1}{6}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{1}{6}\right)^2\right) = \frac{1}{12}$
x^3	$\frac{3!0!}{5!} = \frac{6}{120} = \frac{1}{20}$	$\frac{1}{6}\left(\left(\frac{2}{3}\right)^3 + \left(\frac{1}{6}\right)^3 + \left(\frac{1}{6}\right)^3\right) = \frac{11}{216}$

因此该求积公式的代数精度为 2.

Problem 5

Let $\mathcal{D} = \{(x, y) \in \mathbb{R}^2 : x + y \leq 1, x \geq 0, y \geq 0\}$.

Estimate $\iint_{\mathcal{D}} e^x \sin(y) dx dy$ by partitioning $\mathcal{D}$ with a triangular mesh and applying a composite quadrature rule. Compare your result with the exact one.

Solution:

$$\begin{aligned}
\iint_{\mathcal{D}} e^x \sin(y) dx dy &= \int_0^1 \sin(y) \int_0^{1-y} e^x dx dy \\
&= \int_0^1 \sin(y) [e^{1-y} - 1] dy \\
&= e \int_0^1 \sin(y) e^{-y} dy - \int_0^1 \sin(y) dy
\end{aligned}$$

注意到:

$$\begin{aligned}
\int_0^1 \sin(y) e^{-y} dy &= -\sin(y) e^{-y} \Big|_0^1 + \int_0^1 \cos(y) e^{-y} dy \\
&= -\sin(1) e^{-1} - \cos(y) e^{-y} \Big|_0^1 - \int_0^1 \sin(y) e^{-y} dy \\
&= -\sin(1) e^{-1} - \cos(1) e^{-1} + 1 - \int_0^1 \sin(y) e^{-y} dy \\
&\Downarrow \\
\int_0^1 \sin(y) e^{-y} dy &= \frac{1}{2} [1 - e^{-1} (\sin(1) + \cos(1))]
\end{aligned}$$

于是我们有:

$$\begin{aligned}
 \iint_{\mathcal{D}} e^x \sin(y) dx dy &= e \int_0^1 \sin(y) e^{-y} dy - \int_0^1 \sin(y) dy \\
 &= e \cdot \frac{1}{2} [1 - e^{-1}(\sin(1) + \cos(1))] + \cos(y) \Big|_0^1 \\
 &= \frac{e}{2} - \frac{1}{2} \sin(1) - \frac{1}{2} \cos(1) + \cos(1) - 1 \\
 &= \frac{1}{2} [e - \sin(1) + \cos(1)] - 1
 \end{aligned}$$

Matlab 代码:

```

% Define the original quadrature points
orig_points = [2/3, 1/6;
               1/6, 2/3;
               1/6, 1/6];

% Define the affine transformations for each sub-triangle
% Each transformation now returns a single vector output
transformations = {
    @(p) [0.5*p(1) + 0.5*p(2), 0.5*p(2)];           % Sub1
    @(p) [0.5*p(1) + 0.5, 0.5*p(2)];               % Sub2
    @(p) [0.5*p(1), 0.5*p(2) + 0.5];               % Sub3
    @(p) [0.5*p(2), 0.5*p(1) + 0.5*p(2)];         % Sub4
};

% Initialize the composite estimate
composite_estimate = 0;

% Evaluate and sum contributions from all sub-triangles
for i = 1:4
    T = transformations{i}; % Current transformation
    for j = 1:3
        % Apply transformation to the j-th quadrature point
        p = orig_points(j, :);
        transformed_point = T(p);
        x_trans = transformed_point(1);
        y_trans = transformed_point(2);

        % Evaluate the function at the transformed point
        f_val = exp(x_trans) * sin(y_trans);

        % Accumulate the result with weight 1/24
        composite_estimate = composite_estimate + f_val / 24;
    end
end

% Compute the exact integral for comparison
exact_integral = 0.5*(exp(1) - sin(1) + cos(1)) - 1;

% Display results
fprintf('Composite Estimate: %.6f\n', composite_estimate);
fprintf('Exact Value: %.6f\n', exact_integral);

```

运行结果:

Composite Estimate: 0.208536

Exact Value: 0.208557

Problem 6 (optional)

Use Peano's linear functional representation to analyze the truncation error of the composite trapezoidal rule.

Problem 7 (optional)

Let us ignore all planets in the solar system and assume that the orbit of Halley's comet is a perfect ellipse.

Try to estimate the total orbit length of Halley's comet.

Here are some useful data:

- The eccentricity of the orbit is 0.967.
- The shortest distance between the comet and the sun is 88 million kilometers.

You can also search online to find some other information about the comet.